Warwick School



11+ Entrance Examination Specimen

Mathematics

Please write your full name here:

_____

Before you start read these instructions:

- This test lasts 45 minutes
- We would like to see how you worked out your answers, so show your working. We may be able to give you marks even if the answer is wrong.
- If you get stuck, do not worry. Do not spend lots of time on it, just go on to the next question. You may have time at the end to try the question again.
- Do **not** use a calculator.

Remember to show your working

1. Add $\begin{array}{r} 2837 \\ \underline{\ 248} + \end{array}$ Subtract $\begin{array}{r} 4932 \\ \underline{\ 419} - \end{array}$

Multiply $\begin{array}{r} 284 \\ \underline{\ \ 6} \times \end{array}$ Divide $2863 \div 7$

2. Calculate

$\frac{4}{5} - \frac{1}{3} =$ $\frac{2}{5} + \frac{7}{8} =$

$1\frac{2}{3} \times 3\frac{3}{4} =$ $\frac{5}{6} \div \frac{2}{3} =$

3. Find the next two numbers in these sequences:

a) 3, 7, 11, 15, ... , ...

b) 1, 4, 9, 16, 25, ... , ...

c) 3, 6, 12, 24, ... , ...

4. A piece of wood is 100 centimetres long. It is to be cut into sections that are 8 centimetres long.

a) How many 8 centimetre sections can be cut from the 100 centimetre piece?

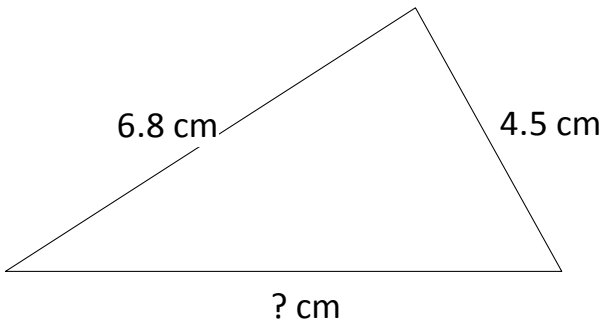
Remember to show your working

b) What length of wood is left over?

5. Work out $263 + 1802 + 493 - 383$

6. Two books together cost £21.99.
One book costs £7.99.
What did the other book cost?

7. The perimeter of this triangle is 18.2 centimetres long. What is the length of the missing side?



8. In old Imperial units there are 12 inches to 1 foot. How many inches are there in 16 feet?

Remember to show your working

9. Subtract 0.0036 from 0.0220

10. A boy measures a length to be 5600 millimetres. What is this in metres?

11. Last night the temperature was $-4^{\circ}C$ but today it is $9^{\circ}C$ warmer. What is today's temperature?

12. Arrange these numbers in order from the smallest to the largest.

¾ 0.72 0.245 70% 0.098

13. A particular type of chocolate bar costs 50 pence. A boy buys *n* chocolate bars that cost a total of *C* pence. Write down a formula for *C* in terms of *n*.

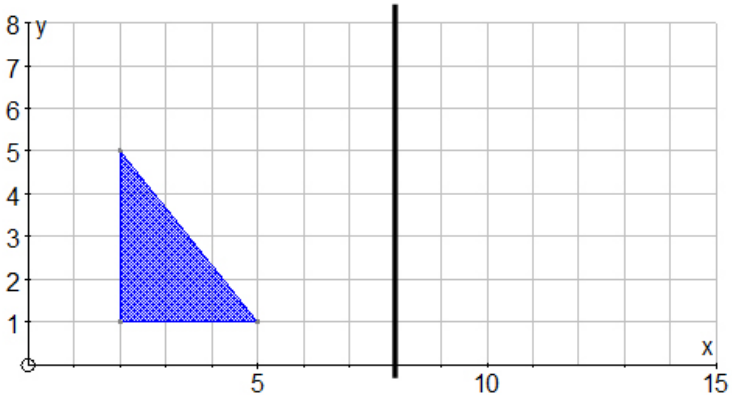
14. A train leaves Warwick station at 15:27 and arrives at London at 17:45. How long did the journey take?

15. What number is half way between 36 and 84?

Remember to show your working

16. I think of a number, double it and add on 12 and the result is 26. What was the number I thought of?

17. a) Write down the co-ordinates of the corners of the triangle.



b) The triangle is reflected in the vertical line. Draw its image on the grid.

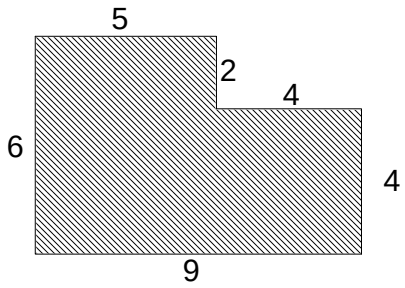
18. Work out 245×34

19. Work out $728 \div 13$

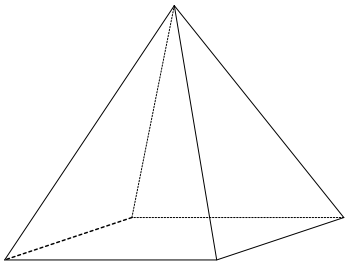
20. A boy collects 'Warhammer' figures. He spends $^{3}/_{5}$ of his pocket money on a Land Raider and $^{1}/_{4}$ on a Captain. What fraction of his pocket money has he got left?

Remember to show your working

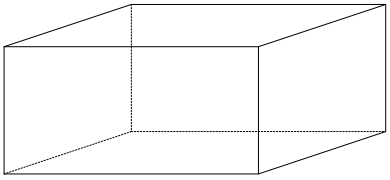
21. Work out the shaded area.



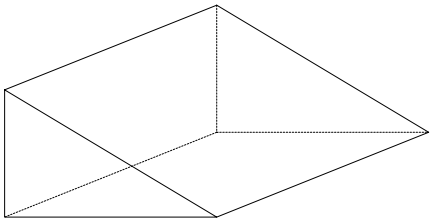
22. Which shape has 5 faces and 9 edges?



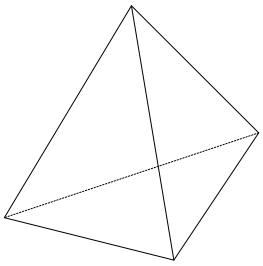
Square based pyramid



Cuboid



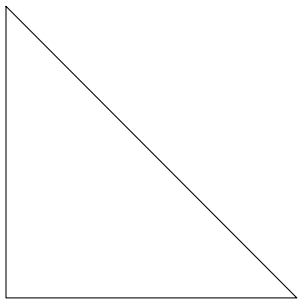
Triangular prism

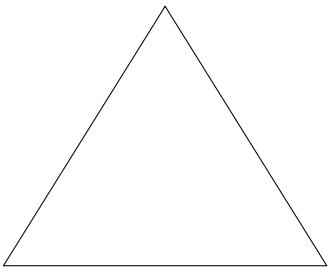


Tetrahedron

Answer:

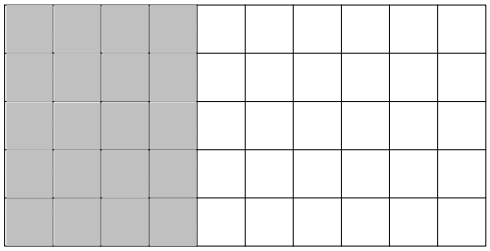
23. Draw the lines of symmetry on these two triangles.





Remember to show your working

24. What percentage of the shape is shaded?



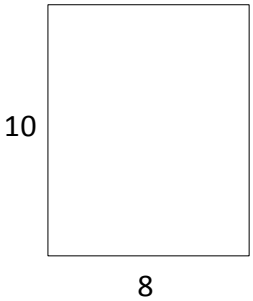
25. A boy draws a plan of his house using a scale of 2 centimetres to 5 metres. He draws the living room to be 5 centimetres long. How long is it in real life?

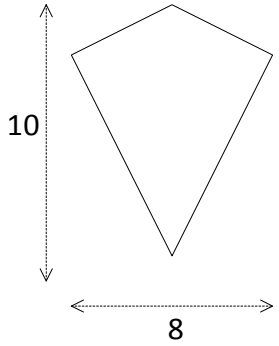
26. Calculate 60% of £30.

27. Write down all the prime factors of 42.

28.

a) What is the area of the rectangle?

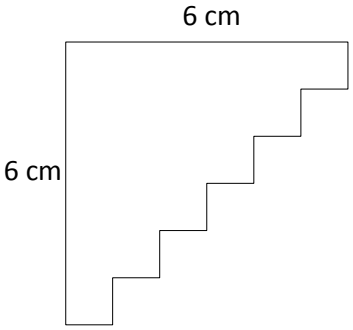




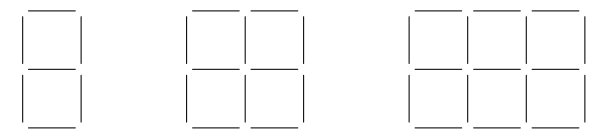
b) What is the area of the kite?

Remember to show your working

29. What is the perimeter of this shape?



30. Here are some matchstick patterns.



Length 1 Length 2 Length 3

a) How many matchsticks are there in a pattern of length 4?

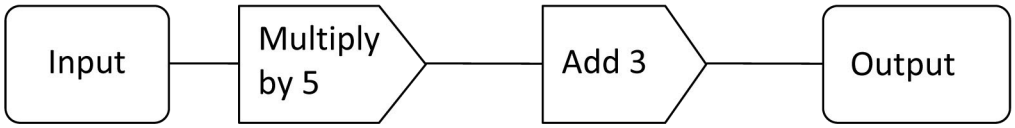
b) How many matchsticks are there in a pattern of length 5?

c) How many matchsticks are there in a pattern of length 10?

31. Work out 999999×456.
(You do not **need** to do a long multiplication sum!)

Remember to show your working

32.



a) If the input is 6, what is the output?

b) If the output is 15, what is the input?

c) If the input is *x*, what is the output?

33. You are given that $12.36 \times 2.94 = 36.34$.

a) What is 1.236×0.00294?

b) What is $36340 \div 1.236$?

END OF EXAMINATION
Now go back and check your answers, and try any questions you may have left out.